6CCSAALA: Advanced Logics for Artificial Intelligence

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5 Introduction to Fuzzy Logics

Learning Objectives

By the end of this section you should be able to:

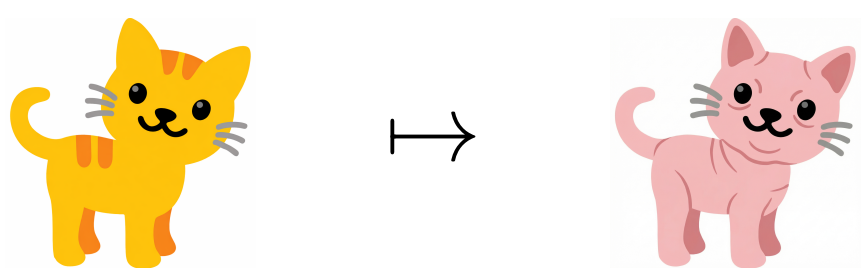
- ☐ Evaluate arbitrary propositional formulas in each of the Fuzzy logics $\mathbb{F}^1$, $\mathbb{F}^2$, and $\mathbb{F}^3$;
- ☐ Demonstrate Trillas' Theorem by constructing an appropriate isomorphism for any given 'reasonable' function for negation;

5.1 Motivation and Definitions

Consider the following argument:

Premise 1)	Mr Whiskers is not bald;
Premise 2)	If Mr Whiskers is not bald and we pluck one hair from Mr Whiskers, then Mr Whiskers would still remains not bald;
Conclusion)	Mr Whisker still remains not bald.

On the face of it this seems like a perfectly natural argument that we might formulate in propositional logic[1]. However, if we were to continue painstaking plucking hairs from Mr Whiskers — in what is doubtless a far more painful experience for him than for us —then we would continue to maintain that 'Mr Whiskers remains not bald' is true even once Mr Whiskers looks like this:



[1]or if we were concerned about the temporal aspect of this, we could formulate this in predicate logic in terms of Mr Whiskers is not bald at time t

How, then, are we to make sense of what is clearly an undesirable conclusion?

Well, upon closer inspection, we may question the validity of a proposition such as Premise 2, and perhaps even if it makes sense to assign a boolean value to a proposition such as 'Mr Whiskers is (not) bald' — surely the issue here is that 'baldness' is not a binary switch where one is either 'bald' or 'not bald' but a sliding scale[2].

Definition 5.1: Boolean Truth Functions

We define a **Boolean truth function** to be a function on n-variables to be a function $F : \{0,1\}^n \to \{0,1\}$ that takes as input an n-tuple of values that are either 0 (false) or 1 (true) and returns a boolean output from $\{0,1\}$.

Each of our Boolean connectives $\neg$, $\wedge$, $\vee$, $\to$, therefore correspond to an equivalent truth function $F_\neg$, $F_\wedge$, $F_\vee$, $F_\to$, of the appropriate arity.

$$\begin{aligned} F_\neg(x) &= 1 - x \\ F_\wedge(x,y) &= \text{min}(x,y) \\ F_\vee(x,y) &= \text{max}(x,y) \\ F_\to(x,y) &= \text{max}(1-x,y) \end{aligned}$$

We can also associate each propositional formula F, with n-variables, with a corresponding truth function $\overline{F} : \{0,1\}^n \to \{0,1\}$ by composing the above functions according to the structure of F. For example, if $F = (p \wedge \neg q) \to r$ then we would have that

$$\overline{((p \wedge \neg q) \to r)}(x_1, x_2, x_3) \;=\; F_\to(F_\wedge(x_1, F_\neg(x_2)), x_3),$$

In general we have that

$$V(F) = 1 \qquad \Longleftrightarrow \qquad \overline{F}\big(V(p_1), \ldots, V(p_n)\big) = 1$$

Effectively, this is just a reformulation of the semantics of classical propositional logic in forms of composition of truth functions. However, with this new formulation,

[2] If we object that 'baldness' is indeed a binary switch and possessing even a single hair is sufficient to be 'not bald', then swap this example for 'gains of sand' in a 'pile', where surely we would contest that a single gain of sand or even half a dozen grains of sand formed a 'pile of sand'

we have the opportunity to relax the condition that our truth functions only take on Boolean values from $\{0, 1\}$, and instead consider truth values over the whole **unit interval** $[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

Definition 5.2: Fuzzy Truth Functions

We define a **Fuzzy** or **real-valued truth function** on n-variables to be a function $F : [0, 1]^n \to [0, 1]$.

A **model** of **fuzzy logic** is a real-valued function $V : \mathsf{Prop} \to [0, 1]$ that assigns a real-value $V(p) \in [0, 1]$ to each propositional variable $p \in \mathsf{Prop}$.

Unlike Boolean truth functions — of which there are at most 2^{2^n} such functions on n-variables — there is seemingly no limit to how creative we could be with our choice of real-valued truth functions on n-variables. However, in practice, when people speak of "Fuzzy logic" they are usually speaking of one of the following semantics:

Min-Max/Zadeh Logic

$$\begin{aligned} F_\neg(x) &= 1 - x \\ F_\wedge(x, y) &= \mathsf{min}(x, y) \\ F_\vee(x, y) &= \mathsf{max}(x, y) \\ F_\to(x, y) &= \mathsf{min}(1, 1 - x + y) \end{aligned}$$

Product/Gödel Logic $\mathbb{F}^2$

$$\begin{aligned} F_\neg(x) &= 1 - x \\ F_\wedge(x, y) &= xy \\ F_\vee(x, y) &= x + y - xy \\ F_\to(x, y) &= \begin{cases} 1 & \text{if } x \leq y \\ y/x & \text{otherwise} \end{cases} \end{aligned}$$

Łukasiewicz Logic $\mathbb{F}^3$

$$
\begin{aligned}
F_{\neg}(x) &= 1 - x \\
F_{\wedge}(x, y) &= \mathsf{max}(0, x + y - 1) \\
F_{\vee}(x, y) &= \mathsf{min}(1, x + y) \\
F_{\rightarrow}(x, y) &= \mathsf{min}(1, 1 - x + y)
\end{aligned}
$$

Exercise

Evaluate each of the following formulas in each of the fuzzy logics $\mathbb{F}^1$, $\mathbb{F}^2$ and $\mathbb{F}^3$ under the valuation $V(p) = 0.3$ and $V(q) = 0.8$:

Formula	**Min–Max $\mathbb{F}^1$**	**Product $\mathbb{F}^2$**	**Łukasiewicz $\mathbb{F}^3$**
$\neg p$			
$p \wedge q$			
$p \vee q$			
$p \rightarrow q$			
$q \rightarrow p$			
$(p \wedge q) \rightarrow p$			
$p \wedge p$			
$p \vee \neg p$			
$p \wedge \neg p$			
$(p \rightarrow q) \vee (q \rightarrow p)$			

Definition 5.3

Given an n-variable function $F(p_1, \ldots, p_n)$ with variables $p_1, \ldots, p_n$. We say that a model $V : \text{Prop} \to [0,1]$ is a **model of** F **with respect to a fuzzy logic** $\mathbb{F}$ if $\overline{F}(V(p_1), \ldots, V(p_n)) = 1$.

We say that F is **valid with respect to** $\mathbb{F}$ or a **theorem of** $\mathbb{F}$ if $\overline{F}(x_1, \ldots, x_n) = 1$ for all $x_1, \ldots, x_n \in [0,1]$. In which case we write

$$\models^{\mathbb{F}} F \qquad \Longleftrightarrow \qquad \overline{F}(x_1, \ldots, x_n) = 1$$

5.2 Interlude 1: The Intermediate Value Theorem

Many of the results in the following sections depend on an important (and almost obvious) property of continuous real-valued functions, called the **Intermediate Value Theorem (I.V.T.)**.

Definition 5.4

A function $f : [a, b] \to \mathbb{R}$ is said to be **continuous at** $c \in [a, b]$ if $\lim_{x \to c} f(x) = f(c)$, *i.e.*

$$\forall \varepsilon > 0;\ \exists \delta > 0;\ |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$$

We say that f is **continuous on** $[a, b]$ (or simply **continuous**) if f is continuous at c for all $c \in [a, b]$.

Theorem 5.5: The Intermediate Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be any continuous real valued function defined on the interval $[a, b]$, and let $N \in \mathbb{R}$ be some value between $f(a)$ and $f(b)$. Then there is some $c \in [a, b]$ such that $f(c) = N$.

Proof: Suppose that $f(a) < N < f(b)$ and let $c = \sup(H)$ where

$$H = \{x \in [a, b] : f(x) \leq N\},$$

which is to say that c is the **least upper bound** of all $x \in [a, b]$ such that $f(x) \leq N$.

Notes

Show that $H \neq \varnothing$ and that $\sup(H)$ is well-defined.

Note that $c \in (a, b)$ since $f(a) < N < f(b)$, by supposition.

Case 1) Suppose that $f(c) < N$ and let $\varepsilon = N - f(c)$, so by continuity there is some $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. Let $d = c + \frac{\delta}{2}$ then $|f(d) - f(c)| < N - f(c)$ from which we have that $f(d) < N$. Hence $d \in H$ and so $c \geq d = c + \frac{\delta}{2}$ since c is an **upper bound** on H, which is a contradiction.

Notes

Sketch a figure illustrating Case 1.

Case 2) Similarly, if $f(c) > N$ then let $\varepsilon = f(c) - N$, so by continuity there is some $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. Let $d = c + \frac{\delta}{2}$ then for all $x \in [d, c]$ it follows that $|f(x) - f(c)| < f(c) - N$, from which we get that $N - f(c) < f(x) - f(c)$, so $f(x) > N$. Which is to say that if $x \in [d, c]$ then $x \notin H$.

Hence $H \subseteq [a, d)$ and so d is an upper bound of H, which is a contradiction since $c > d$ is the **least** upper bound.

Notes

Sketch a figure illustrating Case 2.

Since $<$ is a **total linear order** on $[0, 1]$, we must have that $f(c) = N$, as required.

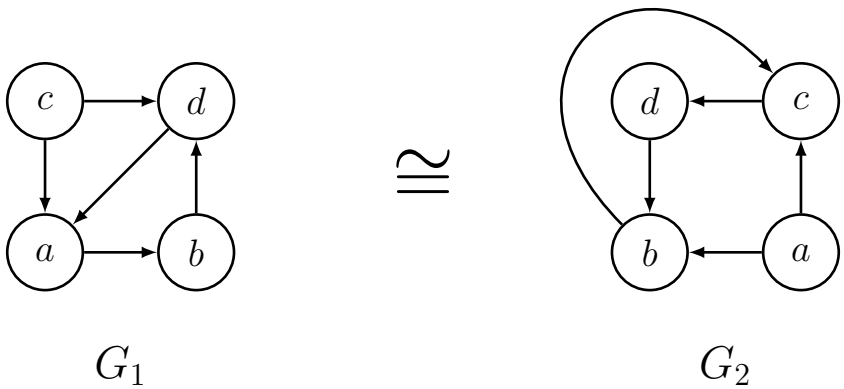
Exercise

Use the Intermediate Value Theorem to show that there are always two antipodal points on Earth (points on opposite sides of the Earth) that have the same temperature.

(You may assume that the Earth is a perfect sphere, and that temperature changes continuously as we navigate around the surface on the Earth.)

5.3 Interlude 2: Isomorphism

We say that two 'structures' are **isomorphic** if they are fundamentally the same up to some relabelling of objects. For example, we have that the following two graphs are isomorphic since we can relabel and rearrange the vertices and edges so that G_1 "looks like" G_2 using the relabelling function that maps $f(a) = c$, $f(b) = d$, $f(c) = a$, $f(d) = b$:



Definition 5.6: Graph Isomorphism

Given two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, we say that G_1 is **isomorphic** to G_2 (denoted $G_1 \cong G_2$) if there exists some bijection $\rho : V_1 \to V_2$ such that

$$(x, y) \in E_1 \quad \iff \quad \big(\rho(x), \rho(y)\big) \in E_2$$

In which case we call ρ an **isomorphism** between G_1 and G_2.

However, we can also define isomorphisms between arbitrary structures; of interest to us will be isomorphisms between **ordered structures**[3] of the form $\langle I, <, f_1, \ldots, f_n, < \rangle$ where I is some interval of $\mathbb{R}$, $<$ is a **total linear ordering** on I and $f_1, \ldots, f_n$ are functions defined on the domain I.

[3] In general, isomorphisms can be defined over arbitrary first-order structures with arbitrarily many predicate symbols, function symbols, and constant symbols.

Definition 5.7: Isomorphism of Ordered Structures

We say that to ordered structures $M_1 = \langle I, <, f \rangle$ and $M_1 = \langle I', <, g \rangle$ are **isomorphic** if there exists some bijection $\rho : I \to I'$ such that:

- **Order is preserved** For all $x, y \in I$ we have that

$$x < y \qquad \Longleftrightarrow \qquad \rho(x) < \rho(y)$$

- **Functions are preserved** For all $x_1, \ldots, x_n, y \in I$, we have that

$$f(x_1, \ldots, x_n) \;=\; y \qquad \Longleftrightarrow \qquad g\big(\rho(x_1), \ldots, \rho(x_n)\big) \;=\; \rho(y)$$

In which case we call ρ an **isomorphism** between M_1 and M_2.

Exercise

Show that the following pairs of ordered structures are isomorphic by construction an appropriate isomorphism:

- $\langle [0,1], <, x^2 \rangle$ and $\langle [0,2], <, x^2/2 \rangle$

- $\langle \mathbb{R}^{>0}, <, \times \rangle$ and $\langle \mathbb{R}, <, + \rangle$

- $\langle [0,1], <, \text{min} \rangle$ and $\langle [-1,1], >, \text{max} \rangle$ (note the change of ordering from $<$ to $>$)

5.4 Trillas' Theorem

One of the first things we notice when looking at the three logics $\mathbb{F}_1$, $\mathbb{F}_2$, and $\mathbb{F}_3$, is that despite the variety of truth functions for $F_\wedge$, $F_\vee$ and $F_\rightarrow$, all three logics are oddly in agreement that $F_\neg(x) = 1 - x$. To answer this question we can start by first introducing some desirable properties that we would like $F_\neg$ to have:

(N1) Boolean: $\neg 0 = 1$ and $\neg 1 = 0$;

(N2) Involutive: $\neg\neg x = x$;

(N3) Continuous: $\neg$ is *continuous*, *i.e.* For all $\varepsilon > 0$ there is a $\delta > 0$ such that $|\neg x - \neg y| < \varepsilon$, whenever $|x - y| < \delta$;

(N4) Decreasing: $F_\neg$ is *decreasing*. *i.e.* If $x < y$ then $\neg x \geq \neg y$;

The first **(N1)** says that negation behaves as we would expect on Boolean inputs. The second **(N2)** says that $F_\neg$ is an **involution**, which also aligns with our expectations, as we would ideally like that double negations cancel each other out — *"It is not the case that Mr Whiskers is not bald"* should be assigned the same truth value as *"Mr Whiskers is bald"*. Continuity **(N3)**, says that if we change the degree to which to assign a truth value to some formula φ changes by some small amount, we should expect that the degree to which we assign a truth value to the negation $\neg\varphi$ also changes by some small amount. And lastly, **(N4)** says that as the degree to which to assign truth to a formula φ increases, then the degree to which we assign truth to its negation $\neg\varphi$ decreases.[4].

[4]Note that **(N4)** only says that $\neg$ is not *increasing*; however, it can be shown that any function satisfying **(N1)–(N4)** must be *strictly* decreasing.

Exercise

Show that if $\neg : [0,1] \to [0,1]$ is a function satisfying **(N2)**, then $\neg$ is a **bijection**.

- **One-to-One (Injective):**

- **Onto (Surjective):**

So is $f(x) = 1 - x$ the only function that has these desirable properties? Surely not!?

Exercise

Write down a function, other than $f(x) = 1 - x$, that satisfies **(N1)**–**(N4)**.

So what gives!?

Well, the following theorem of Trillas (/tɾi.ɟas/) shows that, although there is nothing unique about the function $f(x) = 1 - x$, any function satisfying **(N1)–(N4)** is in, some sense, "equivalent" to $f(x)$ [1].

Theorem 5.8: Trillas' Theorem [1]

If $\neg$ satisfies **(N1)–(N4)**, then $\langle [0,1], <, \neg \rangle$ is isomorphic to $\langle [0,1], <, 1-x \rangle$. That is to say that $\neg x = 1 - x$, up to isomorphism.

Proof:

- First define a function $f(x) = \neg x - x$, and note that, by **(N1)**, $f(0) = \neg 0 - 0 = 1$ and $f(1) = \neg 1 - 1 = -1$. Hence by the **Intermediate Value Theorem**, there is some $c \in [0,1]$ such that $f(c) = \neg c - c = 0$, which is to say that $\neg c = c$.

- Define new function $\rho : [0,1] \to [0,1]$ by taking

$$\rho(x) = \begin{cases} \dfrac{x}{2c} & \text{if } x \leq c \\ 1 - \dfrac{\neg x}{2c} & \text{if } x \geq c \end{cases}$$

- Note that $f(0) = 0$ and $f(1) = 1$

- We claim that $\rho(\neg x) = 1 - \rho(x)$.

  - If $x \leq c$ then $\neg x \geq \neg c$, by **(N4)**,and so by **(N2)** we have that

$$\rho(\neg x) = 1 - \frac{\neg(\neg x)}{2c} = 1 - \frac{x}{2c} = 1 - \rho(x)$$

  - Similarly, if $x > c$ then $\neg x < \neg c$, by **(N4)**, and so

$$\rho(\neg x) = \frac{\neg x}{2c} = 1 - \left(1 - \frac{\neg x}{2c}\right) = 1 - \rho(x)$$

One way to interpret Trillas' Theorem is that although we may disagree on the function $F_\neg$, we can explain these differences as simply a disagreement in the **scale**; that is how we subjectively assign a numerical real-value in $[0,1]$ to our propositions. Of all such reasonable functions satisfying **(N1)–(N4)**, the function $F_\neg(x) = 1 - x$ is the simplest, and so it is the natural choice of truth function for negation that is shared across all fuzzy semantics $\mathbb{F}^1, \mathbb{F}^2, \mathbb{F}^3$

Exercise

Consider the function $\neg : [0,1] \to [0,1]$ with

$$\neg x \;=\; \frac{1-x}{1+x}$$

Show that $\langle [0,1], <, \neg \rangle$ is isomorphic to $\langle [0,1], <, 1-x \rangle$ by constructing an order-preserving isomorphism $\rho : [0,1] \to [0,1]$ such that

$$\rho(\neg x) \;=\; 1 - \rho(x)$$

References

[1] Enric Trillas. Sobre funciones de negación en la teoría de conjuntos difusos. *Stochastica*, 3(1):47–60, 1979.

5.5 Reflections

Reflections

Reflect and summarise the key points from this week's content.